Quiz: noc21_cs17

Q1. What is the primary goal of the second part of the IoT implementation lecture with Raspberry Pi?
(A) To implement a web server on Raspberry Pi.
(B) To integrate a DHT sensor and include networking components for remote data logging.
(C) To develop a mobile application for IoT device control.
(D) To solely focus on local data storage on the Raspberry Pi.

Q2. Which type of sensor is integrated with the Raspberry Pi in this lecture?
(A) PIR sensor
(B) Ultrasonic sensor
(C) DHT sensor
(D) Light sensor

Q3. What networking protocol is used to establish the connection between the Raspberry Pi client and the remote desktop server?
(A) TCP
(B) HTTP
(C) UDP
(D) MQTT

Q4. What is the role of the Raspberry Pi in the client-server model demonstrated?
(A) Server
(B) Router
(C) Client
(D) Gateway

Q5. What is the role of the remote desktop in the client-server model demonstrated?
(A) Client
(B) Sensor
(C) Server
(D) Actuator

Q6. What is the purpose of the 4.7 kOhm resistor when connecting the DHT sensor to the Raspberry Pi?
(A) To limit current to the DHT sensor.
(B) To be obtained between the VCC and the data pin.
(C) To provide power to the Raspberry Pi.
(D) To act as a temperature-sensing element.

Q7. How many pins does the DHT sensor discussed in the lecture typically have, when viewed from left to right with the grid side up?
(A) 2
(B) 3
(C) 4
(D) 5

Q8. Which pin of the DHT sensor is used for transmitting data to the Raspberry Pi?
(A) Pin 1 (VCC)
(B) Pin 2 (Data)
(C) Pin 3 (NC)
(D) Pin 4 (Ground)

Q9. To which Raspberry Pi pin (in BCM mode) is the DHT sensor's data pin connected in the demonstrated setup?
(A) Pin 1 (3.3V)
(B) Pin 4 (5V)
(C) Pin 11
(D) Pin 6 (Ground)

Q10. Which Python library is used to integrate the DHT sensor with Raspberry Pi?
(A) RPi.GPIO
(B) time
(C) adafruit_dht
(D) socket

Q11. What is the primary function of the `socket.socket` function in Python?
(A) To establish a physical connection between two devices.
(B) To create an endpoint for network communication.
(C) To encrypt data before transmission.
(D) To resolve domain names to IP addresses.

Q12. Which `socket_family` is generally used for internet protocols in socket programming?
(A) `AF_UNIX`
(B) `AF_INET`
(C) `AF_BLUETOOTH`
(D) `AF_PACKET`

Q13. For the UDP-based communication demonstrated, what `socket_type` is used?
(A) `SOCK_STREAM`
(B) `SOCK_RAW`
(C) `SOCK_DGRAM`
(D) `SOCK_SEQPACKET`

Q14. In the client-server model for this project, what information is crucial for the client to know about the server?
(A) The server's MAC address and subnet mask.
(B) The server's IP address and port number.
(C) The server's operating system and CPU architecture.
(D) The server's hostname and physical location.

Q15. What mode is specified for the GPIO pins when connecting the DHT sensor to Raspberry Pi pin 11?
(A) BOARD mode
(B) BCM mode
(C) OUT mode
(D) IN mode

Q16. What is the default value for the `protocol` argument in the `socket.socket` function?
(A) 1
(B) -1
(C) 0
(D) 100

Q17. When setting up the server, what function is used to associate the socket with a specific IP address and port?
(A) `listen()`
(B) `connect()`
(C) `bind()`
(D) `accept()`

Q18. What happens when a client successfully connects to the server in the demonstrated setup?
(A) The server immediately closes the connection.
(B) The server sends a 'connection successful' message.
(C) The server requests the client's IP address.
(D) The server starts a new thread for each client.

Q19. What is the purpose of the `while True` loop in the server-side script?
(A) To terminate the server after one client connection.
(B) To continuously listen for and receive data from clients.
(C) To establish a new socket for each incoming connection.
(D) To display a welcome message to new clients.

Q20. How is the received data stored on the server side in the demonstration?
(A) It is printed to the console only.
(B) It is stored in a database.
(C) It is written to a text file in append mode.
(D) It is discarded after being displayed.

Q21. Why is it important for the server to retain a singular IP address, while clients can have a variety of IP addresses, in this one-way communication scenario?
(A) To simplify client-side programming by not requiring them to know the server's IP.
(B) To ensure the server can uniquely identify and log data from each client.
(C) To allow the server to dynamically assign IP addresses to connecting clients.
(D) To prevent unauthorized clients from connecting to the server.

Q22. What happens to the `data_log.txt` file if it doesn't exist when the server starts?
(A) The server will throw an error and terminate.
(B) The server will automatically create the file.
(C) The server will wait for the user to create the file manually.
(D) The server will log data to the console instead.

Q23. What two types of readings are obtained from the DHT sensor in this demonstration?
(A) Light and sound
(B) Pressure and altitude
(C) Humidity and temperature
(D) Motion and distance

Q24. In the client code, how is the sensor data (humidity and temperature) prepared before being sent over the network?
(A) It is encrypted using a secure protocol.
(B) It is converted into a binary format.
(C) It is formatted as a string.
(D) It is stored in a list of integers.

Q25. If multiple Raspberry Pi clients are sending data to the same server, how can the server differentiate between the data sources?
(A) By assigning a unique ID to each client in the server code.
(B) By relying on the IP address and port number received with the data.
(C) By requiring clients to send their MAC addresses.
(D) By using a different port for each client connection.

Q26. What happens to the client's socket connection when the user terminates the client program?
(A) It remains open indefinitely.
(B) It automatically reconnects to the server.
(C) The `close()` function is called to terminate it.
(D) It goes into a suspended state.

Q27. What is the maximum chunk size (in bytes) for receiving data specified in the server code using `sock.recvfrom()`?
(A) 1024
(B) 2048
(C) 4096
(D) 8192

Q28. What is the primary benefit of opening the `data_log.txt` file in append mode (`'a'`)?
(A) To overwrite existing data with new readings.
(B) To read data from the file without modification.
(C) To add new data to the end of the file without deleting previous entries.
(D) To create a new file each time the server starts.

Q29. What was demonstrated to show the real-time change in sensor readings?
(A) Blowing cold air on the sensor.
(B) Placing the sensor in water.
(C) Lighting a match in front of the sensor.
(D) Covering the sensor with a cloth.

Q30. After the client program is terminated, what happens to the data stream on the server side?
(A) It continues to receive old data.
(B) It stops receiving new data.
(C) It automatically restarts the client program.
(D) It logs an error message and shuts down.

Q31. What is the significance of the `GPIO.setwarnings(False)` line in the client's sensor data function?
(A) To disable all GPIO functionality.
(B) To prevent warning messages from being displayed.
(C) To set all GPIO pins to input mode.
(D) To reset GPIO pin configurations.

Q32. In the client code snippet for obtaining sensor data, what GPIO pin numbering mode is explicitly set?
(A) BCM mode
(B) BOARD mode
(C) OUT mode
(D) IN mode

Q33. What is the primary advantage of using UDP over TCP for this specific one-way data logging application?
(A) UDP guarantees delivery of all packets.
(B) UDP provides built-in congestion control.
(C) UDP is connectionless and generally faster for simple data streams.
(D) UDP offers more robust error checking.

Q34. What is the purpose of `socket.gethostname()` when setting up a socket?
(A) To get the IP address of a remote server.
(B) To retrieve the local machine's hostname.
(C) To establish a connection to a specific port.
(D) To determine the network latency.

Q35. If the server is connected to twenty different clients, what information stored in the `data_log.txt` file would help in segregating data from each client?
(A) The timestamp of the data.
(B) The type of sensor data.
(C) The client's IP address and port.
(D) The server's own IP address.

Q36. What is the initial state of the server after it starts and before any client connects?
(A) It immediately starts sending data.
(B) It waits for connections from clients.
(C) It closes the socket.
(D) It displays an error message.

Q37. What is the expected output format of the data received by the server from a single client?
(A) A single numerical value.
(B) Two numerical values (humidity, temperature) separated by a comma.
(C) A JSON object containing sensor data.
(D) A binary string.

Q38. The lecture mentions that 'your client will keep on streaming this data over and over again new values not the old redundant values every time every data pole it will generate a new value'. What mechanism ensures this 'new value' generation?
(A) The client program is restarted for each reading.
(B) The `adafruit_dht.read_retry` function is called repeatedly within a loop.
(C) The sensor itself has an internal memory that updates.
(D) The server requests new data from the client.

Q39. What is the purpose of the `message.encode()` function call before sending data from the client?
(A) To encrypt the message for security.
(B) To convert the string message into bytes.
(C) To compress the message for faster transmission.
(D) To add a timestamp to the message.

Q40. If the client program is terminated, and then restarted, what happens to the server's data reception?
(A) The server must be restarted to accept new connections.
(B) The server will automatically resume receiving data from the client.
(C) The server will log an error because the previous connection was closed.
(D) The server will only receive data if a new port is used.

Q41. What is the general aim of the Internet of Things (IoT) as described in the lecture?
(A) To create isolated, independent devices.
(B) To build an interactive environment of networked devices.
(C) To replace human interaction with machines.
(D) To reduce the number of connected devices.

Q42. What is a key consideration when increasing the number of devices in an IoT system?
(A) Ensuring all devices use the same operating system.
(B) Maintaining system performance and interactivity.
(C) Minimizing the physical size of each device.
(D) Maximizing the power consumption of the system.

Q43. What is the primary function of the `adafruit_dht.read_retry` function?
(A) To configure the DHT sensor's pins.
(B) To read temperature and humidity data from the sensor.
(C) To establish a network connection.
(D) To calibrate the sensor.

Q44. In the context of socket programming, what is the difference between a server and a client?
(A) A server initiates communication, while a client waits for connections.
(B) A server performs tasks requested by a client, while a client requests services.
(C) A server only sends data, while a client only receives data.
(D) A server has a dynamic IP address, while a client has a static IP address.

Q45. What type of file was used to log the sensor data on the server?
(A) CSV file
(B) JSON file
(C) Text file
(D) XML file

Q46. What is the purpose of the `sock.bind()` function on the server side?
(A) To connect to a remote client.
(B) To start listening for incoming connections.
(C) To associate the socket with the server's IP address and a specific port.
(D) To send data to a client.

Q47. What is the final step performed by the client program after sending data and before exiting?
(A) It waits for a server response.
(B) It closes the socket connection.
(C) It reconfigures the sensor.
(D) It logs the data locally.

Q48. If the server is designed to handle multiple clients, why is it important for the server to store the address (IP and port) from which data is received?
(A) To send a confirmation message back to the client.
(B) To identify and segregate data from different clients in logs.
(C) To establish a new TCP connection for each client.
(D) To block specific clients from sending data.

Q49. What is the next topic mentioned for future lectures after this implementation?
(A) Advanced sensor integration.
(B) Data analysis, plotting, and refinement.
(C) Mobile application development for IoT.
(D) Cloud integration for IoT data.

Q50. What is the purpose of the `GPIO.setmode` function in the client's sensor data function?
(A) To set the operating frequency of the sensor.
(B) To define the numbering scheme for GPIO pins.
(C) To initialize the network interface.
(D) To calibrate the sensor readings.

Q51. What is the primary focus of the third part of the IoT implementation lecture with Raspberry Pi?
(A) Integrating new types of sensors with Raspberry Pi.
(B) Configuring Raspberry Pi as a server device.
(C) Server-side data processing and visualization.
(D) Optimizing client-side data transmission protocols.

Q52. In the demonstrated system overview, what is the role of the Raspberry Pi?
(A) To act as a remote server for data storage.
(B) To connect to sensors, read data, and send it to a remote server.
(C) To process and visualize data locally.
(D) To host the running plot of incoming data.

Q53. For the DHT sensor connected to the Raspberry Pi, which pin of the DHT sensor is connected to the data pin of the Raspberry Pi?
(A) Pin 1
(B) Pin 4
(C) Pin 2
(D) Pin 11

Q54. Why is data processing, such as filtering, necessary when receiving data from multiple IoT sensors on a remote server?
(A) To reduce the overall volume of data stored on the server.
(B) To ensure data is properly formatted and handle corruption or incompleteness.
(C) To encrypt the data for secure transmission.
(D) To convert data into a different unit of measurement.

Q55. If a single IoT node sends multiple sensor readings (e.g., temperature, humidity, light intensity) to a server, why would data splitting be required on the server side?
(A) To compress the data for efficient storage.
(B) To allow individual access to specific sensor values.
(C) To combine data from different nodes into a single record.
(D) To convert the data into a graphical format.

Q56. In Python, which built-in function is used to split a string into multiple substrings based on a specified delimiter?
(A) divide()
(B) separate()
(C) split()
(D) parse()

Q57. If a data string is 'sunday,Monday,tuesday' and you want to split it into individual day names, what would be the delimiter used with the split() function?
(A) 'sunday'
(B) ','
(C) 'Monday'
(D) 'tuesday'

Q58. Which Python library is primarily used for plotting data, similar to its use in MATLAB?
(A) numpy
(B) pandas
(C) matplotlib
(D) scipy

Q59. To plot data with matplotlib, which command is used to define the x and y axis values for a basic 2D plot?
(A) xlabel(x_values)
(B) ylabel(y_values)
(C) plot(x, y)
(D) title(plot_title)

Q60. To add a label to the y-axis of a plot using matplotlib, which function would you use?
(A) plt.xaxis_label()
(B) plt.y_label()
(C) plt.ylabel()
(D) plt.set_y_label()

Q61. Which matplotlib function is generally used for animations and interactive plot modes?
(A) plt.figure()
(B) plt.grid()
(C) plt.ion()
(D) plt.show()

Q62. What is the name of the new function added on the server side to animatedly plot incoming data?
(A) data_plot()
(B) coverage_plot()
(C) server_plot()
(D) animate_data()

Q63. In the coverage_plot function, after splitting the incoming data (which was in CSV format), what does the zeroth element represent?
(A) Temperature
(B) Humidity
(C) Iteration number
(D) Sensor ID

Q64. Which matplotlib command is used in the coverage_plot function to redraw the image without overwriting previous parts, enabling an animated plot?
(A) plt.show()
(B) plt.close()
(C) fig.canvas.draw()
(D) plt.update()

Q65. Why does the client device (Raspberry Pi) not require an IP address in the described one-way communication setup?
(A) It only sends data, it does not receive.
(B) It uses a dynamic IP address.
(C) The server assigns it an IP address.
(D) It communicates via Bluetooth, not IP.

Q66. What is a significant advantage of designing the IoT architecture such that changes can be made primarily on the server side?
(A) It reduces the power consumption of client devices.
(B) It allows for easier and more efficient updates across multiple clients.
(C) It enhances the security of data transmission.
(D) It enables clients to process data locally.

Q67. The lecture mentions that incoming data from the client is saved into a text file. What common format is used when values are separated by a comma?
(A) XML
(B) JSON
(C) CSV
(D) TXT

Q68. Besides a comma, what other characters are mentioned as possible delimiters for splitting data?
(A) Only spaces
(B) Semicolon, colon, or even a question mark
(C) Only tabs
(D) Hyphen and underscore

Q69. When data.split() is called on a string with a specified delimiter, what is the typical output?
(A) A single string with the delimiter removed.
(B) A list of individual substrings.
(C) A boolean indicating if the split was successful.
(D) An integer representing the number of splits.

Q70. What is the purpose of the title() function in matplotlib?
(A) To set the title of the x-axis.
(B) To set the title of the y-axis.
(C) To give a title to the entire plot.
(D) To name the figure window.

Q71. When plt.plot(a) is called with only one argument 'a' (a range of values), what does matplotlib automatically assume for the x-axis?
(A) It uses the index of the values in 'a'.
(B) It uses a random set of numbers.
(C) It uses the same values as 'a'.
(D) It prompts the user for x-axis values.

Q72. Which library is imported as np and used to generate random numbers for the y-axis in the scatter plot example?
(A) random
(B) math
(C) numpy
(D) scipy

Q73. Which matplotlib command is used to display a grid on a plot?
(A) plt.show_grid()
(B) plt.grid(True)
(C) plt.add_grid()
(D) plt.display_grid()

Q74. On the server side, what is required for socket creation to listen for incoming client connections?
(A) Only the server's IP address.
(B) Only the port number.
(C) The server's IP address and the particular port it is listening on.
(D) The client's IP address and port.

Q75. In the server-side code, how many bytes of data are received in chunks from the client?
(A) 1024 bytes
(B) 2048 bytes
(C) 4096 bytes
(D) 8192 bytes

Q76. In the coverage_plot function, why is plt.figure(2) used instead of plt.figure() without an argument?
(A) To create a new figure window each time.
(B) To ensure the plot is displayed in a specific, existing figure window.
(C) To set the size of the figure.
(D) To make the plot interactive.

Q77. How is the data packaged and transmitted from the client (Raspberry Pi) to the server?
(A) As a JSON object.
(B) In the form of h (humidity) and t (temperature).
(C) As a raw binary stream.
(D) As an XML document.

Q78. Beyond visualizing sensor data, what other capability is mentioned if the system is hosted on the internet?
(A) Local data processing only.
(B) Remotely actuating sensors or IoT nodes.
(C) Changing the sensor type.
(D) Disconnecting from the network.

Q79. What is the main goal of the overall system described in the lecture?
(A) To create a standalone data logger.
(B) To develop an interactive environment using a network of connected devices.
(C) To replace physical sensors with virtual ones.
(D) To perform complex mathematical simulations.

Q80. Where can data logging and processing occur in the described IoT system?
(A) Only on the client side.
(B) Only on the server side.
(C) On both the client side and the server side.
(D) Only in the cloud.

Q81. What is the function of Pin 1 on the DHT sensor in the described setup?
(A) Data output
(B) Ground
(C) Power
(D) Not connected

Q82. What is the function of Pin 4 on the DHT sensor in the described setup?
(A) Data output
(B) Ground
(C) Power
(D) Not connected

Q83. Which specific Python library is mentioned for interfacing with the DHT sensor?
(A) pyserial
(B) Adafruit Python library
(C) RPi.GPIO
(D) smbus

Q84. Where is the iotsr.py file, which generates temperature and humidity values, kept?
(A) On the server desktop.
(B) On the Raspberry Pi (client device).
(C) In a cloud storage.
(D) On a separate data logger.

Q85. When the Raspberry Pi is connected to the network and communicates with a desktop acting as a server, what kind of architecture is established?
(A) Peer-to-peer architecture.
(B) Mesh network architecture.
(C) Client-server architecture.
(D) Broadcast architecture.

Q86. Which of the following is NOT mentioned as a reason for potential data corruption or incompleteness when many sensors upload data rapidly?
(A) Server not able to receive the whole data.
(B) Incomplete data being transmitted.
(C) Client sending too much data.
(D) Data not being properly formatted.

Q87. If the incoming data from the client is in the format 'string h,string t', how would you split these values on the server to get individual h and t values?
(A) By using a space as a delimiter.
(B) By using the split() function with a comma as the delimiter.
(C) By using a colon as a delimiter.
(D) By using a newline character as a delimiter.

Q88. What is the primary purpose of the figure() function in matplotlib?
(A) To save the plot to a file.
(B) To create a new figure (window) for plotting.
(C) To display the plot on the screen.
(D) To add a title to the plot.

Q89. What does the subplot() function do in matplotlib?
(A) It closes the current figure.
(B) It adds a subplot within the current figure.
(C) It creates a scatter plot.
(D) It enables interactive mode.

Q90. Which matplotlib function is used to create a scatter point plot of given points?
(A) plot()
(B) line()
(C) scatter()
(D) points()

Q91. The server-side code includes a 'while it is true' loop. What is continuously happening inside this loop?
(A) The server is sending data to the client.
(B) The server is receiving data, storing it, and calling the plotting function.
(C) The server is closing connections.
(D) The server is waiting for user input.

Q92. When does the iterative plotting function on the server side stop?
(A) When the server runs out of memory.
(B) When the server is closed by the user or the client is closed.
(C) After a fixed number of iterations.
(D) When the network connection is lost.

Q93. After splitting the incoming data (e.g., 'humidity,temperature') in the coverage_plot function, what does the first element (index 0) represent and what does the second element (index 1) represent?
(A) Index 0: Temperature, Index 1: Humidity
(B) Index 0: Humidity, Index 1: Temperature
(C) Index 0: Iteration, Index 1: Temperature
(D) Index 0: Temperature, Index 1: Iteration

Q94. In the coverage_plot function, what are the axes for the temperature plot?
(A) Temperature on X-axis, Humidity on Y-axis.
(B) Temperature on Y-axis, Iteration on X-axis.
(C) Iteration on Y-axis, Temperature on X-axis.
(D) Time on X-axis, Temperature on Y-axis.

Q95. If you have 100 clients deployed, and you need to make minor adjustments to the system's functionality, what is the benefit of having a centralized server script?
(A) Each client needs to be updated individually.
(B) Only the server script needs to be updated, saving time and effort.
(C) The clients will automatically update themselves.
(D) The system will continue to function without any updates.

Q96. When iotsr.py is run on the Raspberry Pi, before connecting to the network, what is the Raspberry Pi considered?
(A) A client device.
(B) A server device.
(C) A sensor device.
(D) A network router.

Q97. If the server hosting the running plot is hosted on the internet, what is a key benefit mentioned?
(A) Reduced data storage requirements.
(B) Ability to access sensor readings from any part of the world.
(C) Enhanced local processing capabilities.
(D) Elimination of the need for sensors.

Q98. Once data transmission from the client initiates, where does the successive data get stored on the server?
(A) In a temporary buffer.
(B) In a log file.
(C) In the server's RAM.
(D) It is immediately discarded after plotting.

Q99. What is the role of plt.show() in a basic matplotlib plotting script?
(A) To save the plot to a file.
(B) To close the plot window.
(C) To display the generated plot on the screen.
(D) To clear the plot.

Q100. During the demonstration, when the temperature was increased, what was observed on the plot?
(A) Only the humidity plot changed.
(B) The temperature points on the plot started rising.
(C) Both plots remained constant.
(D) The plot disappeared.

Q101. What is the primary goal of Software Defined Networking (SDN) as described in the lecture?
(A) To increase the physical size of network switches.
(B) To transform or restructure existing network infrastructure for efficiency.
(C) To eliminate the need for any network protocols.
(D) To make all network devices proprietary.

Q102. In traditional networks, what is a key limitation regarding the global view of the network?
(A) Each switch has a complete global view of the entire network.
(B) Switches only have a local view and forward traffic in a distributed manner.
(C) Only the application layer has a global view.
(D) Global view is only available in wireless networks.

Q103. Which of the following is NOT a limitation of existing networks mentioned in the lecture?
(A) Vendor-specific architecture of switches.
(B) Difficulty in dynamic configuration based on application requirements.
(C) Presence of a centralized controller for network management.
(D) Each switch running its own operating system and applications.

Q104. What is the core idea behind SDN to overcome the limitations of traditional networks?
(A) To integrate all applications and operating systems directly into the hardware.
(B) To separate the application and operating system from the hardware in each switch.
(C) To use only proprietary hardware for all network functions.
(D) To eliminate the need for any operating system in network devices.

Q105. Which Stanford University project in 2006 is mentioned as a precursor to SDN, focusing on centralized security policy control?
(A) OpenFlow
(B) Nicira
(C) SANE
(D) Open Networking Foundation

Q106. When did the idea of Software Defined Network originate with the OpenFlow project?
(A) 2006
(B) 2008
(C) 2009
(D) 2011

Q107. What is the primary function of the data plane in the SDN architecture?
(A) Performing security and routing applications.
(B) Defining control logic in a centralized manner.
(C) Forwarding packets.
(D) Managing the network operating system.

Q108. The separation of the data plane from the control plane in SDN is made possible through what process?
(A) Hardware acceleration.
(B) Network Function Virtualization (NFV).
(C) Proprietary software development.
(D) Distributed ledger technology.

Q109. What is the name of the API that facilitates communication between the control plane and the application plane in SDN?
(A) Southbound API
(B) Eastbound API
(C) Northbound API
(D) Westbound API

Q110. Which protocol is currently supporting the southbound API in SDN?
(A) OSPF
(B) TCP/IP
(C) OpenFlow
(D) HTTP

Q111. What are the two main challenges when implementing SDN in any network, as highlighted in the lecture?
(A) Hardware cost and software compatibility.
(B) Rule placement problem and controller placement problem.
(C) Network latency and bandwidth limitations.
(D) User adoption and security vulnerabilities.

Q112. In the context of SDN, what do flow tables replace from traditional networking?
(A) ARP tables.
(B) DNS records.
(C) Routing tables.
(D) MAC address tables.

Q113. Where are flow tables primarily stored in SDN switches?
(A) Hard disk drives.
(B) Standard DRAM.
(C) Ternary Content Addressable Memory (TCAM).
(D) Solid State Drives (SSDs).

Q114. What is a characteristic of TCAM memory that makes it suitable for flow tables but also a challenge?
(A) It is very slow and inexpensive.
(B) It is very fast but limited in size and expensive.
(C) It is unlimited in size and very cheap.
(D) It is only used for temporary storage.

Q115. What message does a switch send to the controller when it receives a packet for which no flow rule is present?
(A) Packet-out message.
(B) Flow-request message.
(C) Packet-in message.
(D) Rule-query message.

Q116. Approximately how much delay is involved in the placement of a new flow rule in a switch after a packet-in message is sent?
(A) Less than 1 millisecond.
(B) 3 to 5 milliseconds.
(C) 10 to 15 milliseconds.
(D) Over 1 second.

Q117. Which of the following is a challenge related to rule placement in SDN?
(A) How to increase the physical size of switches.
(B) How to define rules to minimize packet-in messages to the controller.
(C) How to make TCAM memory slower but cheaper.
(D) How to eliminate the need for any flow rules.

Q118. In OpenFlow, are the fields for matching packets (e.g., source IP, destination IP) fixed or customizable?
(A) They are fully customizable by the network administrator.
(B) They are fixed within a particular version of OpenFlow.
(C) They change dynamically based on network traffic.
(D) They are determined by the application layer.

Q119. What is the difference between a hard timeout and a soft timeout for flow rules?
(A) Hard timeout deletes rules based on inactivity, soft timeout deletes all rules at once.
(B) Hard timeout deletes all rules (e.g., switch reset), soft timeout deletes rules based on inactivity.
(C) Hard timeout is for critical rules, soft timeout is for non-critical rules.
(D) Hard timeout is set by the controller, soft timeout is set by the switch.

Q120. What is the relationship between SDN and OpenFlow?
(A) SDN and OpenFlow are interchangeable terms for the same technology.
(B) SDN is a protocol, and OpenFlow is a concept.
(C) SDN is a technology/concept, and OpenFlow is a protocol used for communication between data and control planes.
(D) OpenFlow is a proprietary implementation of SDN.

Q121. Which of the following is an example of OpenFlow switch software mentioned in the lecture?
(A) Cisco IOS
(B) Junos OS
(C) Open vSwitch
(D) Windows Server

Q122. In traditional networks, if a switch goes down, why is finding an alternate route difficult?
(A) Because all switches have a global view and can easily reconfigure.
(B) Because there is no centralized control over the network.
(C) Because routing protocols like OSPF are too efficient.
(D) Because users are directly connected to the internet.

Q123. What does the lecture imply about the cost and complexity of individual switches in current networks?
(A) They are inexpensive and simple, requiring minimal resources.
(B) They are cost expensive, run thousands of lines of code, and require significant RAM.
(C) They are only used for basic forwarding, not complex applications.
(D) They are becoming obsolete due to cloud computing.

Q124. What is the role of the centralized controller in SDN?
(A) To physically forward packets between switches.
(B) To define the control logic and manage flow rules.
(C) To run user applications directly.
(D) To replace all hardware components in the network.

Q125. Which of the following is NOT considered a component or attribute of SDN?
(A) Hardware switches.
(B) Controller.
(C) Proprietary operating systems for each switch.
(D) Flow rules.

Q126. How does SDN aim to minimize operation cost and delay caused by link failures?
(A) By eliminating all network links.
(B) By using a centralized control to quickly reconfigure routes.
(C) By making all switches self-healing.
(D) By increasing the number of physical cables.

Q127. What is the significance of the 'priority' field in an OpenFlow rule?
(A) It indicates the importance of the packet's content.
(B) It determines the order in which rules are evaluated.
(C) It specifies the physical port for forwarding.
(D) It defines the maximum packet size.

Q128. What does 'match' in an OpenFlow rule refer to?
(A) The action to be taken on a packet.
(B) The specific fields in a packet header that are compared against the rule.
(C) The destination port of the packet.
(D) The time-to-live value of the packet.

Q129. What is the 'instruction' or 'action' in an OpenFlow rule?
(A) The source IP address of the packet.
(B) The command to be executed if a packet matches the rule.
(C) The priority level of the rule.
(D) The version of the OpenFlow protocol.

Q130. Why is it challenging to scale up TCAM memory for flow tables?
(A) Because it is too slow for network operations.
(B) Because it is very expensive.
(C) Because it is incompatible with modern processors.
(D) Because it consumes too much power.

Q131. What is one of the problems researchers are currently working on regarding rule placement?
(A) How to make TCAM memory infinitely large.
(B) How to eliminate the need for any flow rules.
(C) How to define rules to send more packet-in messages to the controller.
(D) How to define or place rules at switches considering limited TCAM availability.

Q132. What does the term 'L3 switch' refer to in the context of the lecture?
(A) A Layer 1 switch.
(B) A Layer 2 switch.
(C) A Layer 3 switch, which is used interchangeably with 'router'.
(D) A switch that only handles local area network traffic.

Q133. What protocol is mentioned as being used for switching or routing in current day networks before SDN?
(A) HTTP
(B) OSPF
(C) FTP
(D) SMTP

Q134. What is the primary characteristic of how switches forward traffic in traditional networks?
(A) They forward traffic based on a global network map.
(B) They forward traffic in a distributed manner without a global view.
(C) They forward traffic only after consulting a centralized controller.
(D) They forward traffic based on application-specific requirements.

Q135. What is the purpose of Network Function Virtualization (NFV) in enabling SDN?
(A) To replace all physical network hardware with software.
(B) To provide a logical view of the network for performing functions like security and routing.
(C) To increase the processing power of individual switches.
(D) To standardize hardware across all vendors.

Q136. Which of the following is an example of an application functionality that runs on the application plane in SDN?
(A) Packet forwarding.
(B) Network operating system.
(C) Security.
(D) TCAM memory management.

Q137. What does the term 'clean slate security architecture' refer to in the context of the SANE project?
(A) An architecture that cleans up old security vulnerabilities.
(B) A completely new security architecture designed from scratch.
(C) A security architecture that only uses open-source components.
(D) An architecture that focuses on physical security only.

Q138. What was founded in March 2011 related to SDN?
(A) The OpenFlow project.
(B) Nicira network company.
(C) The Open Network Foundation.
(D) The first Open Networking Summit.

Q139. Why is it mentioned that changing the current network to SDN must be done in a phased manner?
(A) Because SDN is still an experimental technology.
(B) Because it requires architectural changes and hardware/software support.
(C) Because it is too expensive to implement all at once.
(D) Because users need time to adapt to new interfaces.

Q140. What is the primary difference between the 'applications' mentioned in the context of a switch's stack (hardware, OS, app) and the 'application plane' in SDN?
(A) They are the same; the terms are interchangeable.
(B) Switch applications are user-facing, while application plane applications are network-facing.
(C) Switch applications are tightly coupled with hardware/OS, while application plane applications are decoupled and centralized.
(D) Switch applications are for routing only, while application plane applications are for security only.

Q141. What does the 'star' symbol (e.g., in 'match maybe in a star means that you know it is it it doesnt require any match') in an OpenFlow rule example signify?
(A) It indicates a high priority rule.
(B) It means the field must exactly match a specific value.
(C) It means the field does not require any specific match (wildcard).
(D) It denotes an optional field.

Q142. Which of the following is a characteristic of the OpenFlow protocol regarding its versions?
(A) All versions of OpenFlow use the same fixed set of matching fields.
(B) OpenFlow has only one version, which is constantly updated.
(C) OpenFlow has different versions, and each version has a fixed set of matching fields.
(D) OpenFlow versions are entirely customizable by users.

Q143. What is the primary benefit of having a centralized controller in SDN?
(A) It allows each switch to operate independently.
(B) It provides a global view of the network and centralized control logic.
(C) It reduces the need for any network protocols.
(D) It makes network hardware more expensive.

Q144. What is the role of the 'network operating system' within the control plane of SDN?
(A) To manage the physical hardware of individual switches.
(B) To provide a platform for network-wide control and management.
(C) To run user applications directly on the controller.
(D) To forward packets at high speed.

Q145. Why is SDN considered to have 'lot of potential for use in internet of things'?
(A) Because IoT devices do not require network connectivity.
(B) Because SDN can make IoT networks more efficient.
(C) Because SDN is only compatible with wireless IoT devices.
(D) Because IoT devices are too powerful for traditional networks.

Q146. What is the main characteristic of the 'stack' (hardware, OS, app) running on each switch in traditional networks?
(A) It is highly modular and easily reconfigurable.
(B) It is integrated, with app and OS tightly coupled to hardware.
(C) It is managed by a central network operating system.
(D) It is designed for cloud-native applications.

Q147. Which of the following companies is mentioned as having announced their intention to incorporate SDN into their switches?
(A) Google
(B) Microsoft
(C) Juniper
(D) Amazon

Q148. What is the primary function of flow rules in SDN?
(A) To define the physical layout of the network.
(B) To determine how streams of packets are handled by switches.
(C) To manage user authentication and authorization.
(D) To monitor network bandwidth usage.

Q149. What is the 'controller placement problem' in SDN?
(A) The challenge of physically installing controllers in data centers.
(B) The challenge of deciding where to locate controllers for optimal performance and resilience.
(C) The problem of connecting controllers to the internet.
(D) The issue of controllers being too expensive to deploy.

Q150. What is the main advantage of SDN's centralized control over traditional distributed routing?
(A) It makes network configuration more complex.
(B) It allows for dynamic and flexible network management.
(C) It increases the cost of network infrastructure.
(D) It eliminates the need for any network protocols.

Q151. What is the primary protocol used for the southbound API communication in SDN?
(A) HTTP
(B) OpenFlow
(C) FTP
(D) SNMP

Q152. Which API in SDN is used for communication between the control plane and the application plane?
(A) Southbound API
(B) East-Westbound API
(C) Northbound API
(D) Inter-plane API

Q153. The concept of an East-Westbound API comes into play when discussing what aspect of SDN?
(A) Communication between data plane and control plane.
(B) Communication between control plane and application plane.
(C) The use of multiple controllers in the control layer.
(D) The physical connection between switches and controllers.

Q154. What is the main function of controllers in an SDN environment?
(A) Forwarding data packets based on destination IP.
(B) Defining flow rules according to application-specific requirements.
(C) Managing physical network infrastructure.
(D) Encrypting data for secure transmission.

Q155. What is the typical processing capacity of a single-threaded controller in terms of requests per second?
(A) Approximately 50 requests per second.
(B) Approximately 100 requests per second.
(C) Approximately 200 requests per second.
(D) Approximately 500 requests per second.

Q156. If a large network has a small number of controllers, what is a potential consequence?
(A) Increased data plane efficiency.
(B) Reduced control packet overhead.
(C) Network congestion with control packets.
(D) Faster rule placement.

Q157. How do switches logically perceive their connection distance to a controller in SDN?
(A) Multiple hops away.
(B) Directly connected (zero hops).
(C) One hop distance.
(D) Variable, depending on network size.

Q158. In which controller architecture is a particular switch connected to only one controller, even if multiple controllers are arranged in a specific pattern?
(A) Flat architecture
(B) Hierarchical architecture
(C) Ring architecture
(D) Mesh architecture

Q159. Which controller architecture is known for increasing reliability and improving fault tolerance in an SDN network?
(A) Flat architecture
(B) Hierarchical architecture
(C) Ring architecture
(D) Mesh architecture

Q160. When a network is controlled by a single controller, what type of control mechanism is being used?
(A) Distributed control
(B) Centralized control
(C) Hierarchical control
(D) Federated control

Q161. What is the role of a backup controller in an SDN environment?
(A) To handle all incoming requests when the primary controller is busy.
(B) To provide an alternative path for data traffic.
(C) To take over control of the network if the primary controller fails.
(D) To manage security policies independently.

Q162. According to the example provided, what is the correct sequence of policy enforcement for an HTTP request in an SDN security chain?
(A) HTTP -> Firewall -> IDS -> Proxy -> Switch
(B) HTTP -> Switch -> IDS -> Proxy -> Firewall -> Switch
(C) HTTP -> Proxy -> IDS -> Firewall -> Switch
(D) HTTP -> Switch -> Firewall -> Proxy -> IDS -> Switch

Q163. Which software is specifically mentioned as being very useful for creating a virtual network with OpenFlow enabled switches for experimentation?
(A) Wireshark
(B) Mininet
(C) GNS3
(D) Packet Tracer

Q164. Mininet supports which types of controllers for virtual network experimentation?
(A) Only local controllers.
(B) Only remote controllers.
(C) Both remote and local controllers.
(D) Neither remote nor local controllers.

Q165. Which two controller configuration software options are highlighted as the most popular ones currently used?
(A) POX and NOX
(B) Floodlight and POX
(C) Open Daylight and ONOS
(D) NOX and Open Daylight

Q166. What is a key distinction between an emulator and a simulator in the context of SDN experimentation?
(A) Simulators can connect to real networks, while emulators cannot.
(B) Emulators can be connected to real networks with real traffic, unlike simulators which simulate everything.
(C) Simulators are used for virtual networks, while emulators are for physical networks.
(D) Emulators are typically written in Python, while simulators are not.

Q167. What is the primary motivation behind separating the control logic from the data logic in SDN?
(A) To increase network security.
(B) To make the network more efficient.
(C) To reduce the cost of network hardware.
(D) To simplify network cabling.

Q168. What is a significant trade-off or challenge associated with separating control logic in SDN?
(A) Decreased network latency.
(B) Reduced need for specialized hardware.
(C) Increased control message overhead due to additional packets.
(D) Simplified network management.

Q169. What is the typical time taken for rule placement in a switch's flow table if the rule is not already existing?
(A) Less than 1 millisecond.
(B) Approximately 3 to 5 milliseconds.
(C) Approximately 10 to 15 milliseconds.
(D) More than 1 second.

Q170. The primary goal regarding delay in controller placement is to:
(A) Eliminate all control-linked delay.
(B) Maximize control-linked delay for stability.
(C) Minimize control-linked delay.
(D) Keep control-linked delay constant regardless of network size.

Q171. In SDN, the control logic is primarily handled by which component?
(A) The switches in the data plane.
(B) The application plane.
(C) The control plane (controller).
(D) The network interface cards.

Q172. The southbound API in SDN facilitates communication between which two layers?
(A) Application layer and control layer.
(B) Control layer and data layer.
(C) Application layer and data layer.
(D) Multiple controllers.

Q173. What kind of APIs can be used for the northbound API communication in SDN?
(A) Only OpenFlow protocol.
(B) Only proprietary APIs.
(C) All existing standard APIs.
(D) Only East-Westbound APIs.

Q174. What does the controller know about the network flows in SDN?
(A) Only the destination IP addresses.
(B) How the different flows and packets are going to be handled.
(C) Only the physical topology of the network.
(D) Only the current bandwidth utilization.

Q175. In a flat SDN architecture, what is the sequence of events when a switch receives a flow for which it doesn't have a rule?
(A) Switch sends flow rule to controller -> Controller processes flow.
(B) Switch sends packet-in message to controller -> Controller sends flow rule back to switch.
(C) Switch forwards packet to default gateway -> Gateway sends rule to controller.
(D) Switch drops packet -> Controller detects dropped packet and sends rule.

Q176. In a distributed control mechanism for SDN, how are control decisions typically made?
(A) A single central controller makes all decisions.
(B) Each sub-network is controlled by a different controller.
(C) Decisions are made by the data plane switches.
(D) Decisions are made by the application layer.

Q177. SDN can provide enhanced levels of security by taking help from which network functions?
(A) Only basic routing protocols.
(B) Firewall, proxy, HTTP, and IDS.
(C) Only data encryption algorithms.
(D) Physical access control systems.

Q178. What is the main purpose of using simulators and emulators in SDN?
(A) To deploy production networks.
(B) To conduct real-world traffic analysis only.
(C) To experiment and evaluate performance.
(D) To replace physical network hardware.

Q179. Mininet, a software for switch deployment, runs on which programming language?
(A) Java
(B) C++
(C) Python
(D) Ruby

Q180. What is a desired outcome of SDN in terms of network management?
(A) Decentralized network management.
(B) Isolated network segment management.
(C) Unified network management leveraging a global view.
(D) Manual configuration of each network device.

Q181. The rule placement problem in SDN involves a trade-off between TCAM limitations and what other factor?
(A) Network throughput.
(B) Power consumption.
(C) Delay.
(D) Number of physical ports.

Q182. What is the 'controller placement problem' primarily concerned with?
(A) The physical location of data plane switches.
(B) Optimizing the number and location of controllers in the network.
(C) Deciding which applications can use the northbound API.
(D) Managing the flow of data packets between switches.

Q183. What must a controller be able to handle effectively to ensure proper network operation?
(A) Only outbound data traffic.
(B) All incoming requests from the switches.
(C) Only application-layer protocols.
(D) Physical layer signaling.

Q184. What happens if rule placement incurs too much delay in an SDN network?
(A) Data plane becomes more efficient.
(B) Control overhead increases significantly.
(C) Network security is enhanced.
(D) The network becomes fully decentralized.

Q185. While single-threaded controllers have a capacity limit, what is mentioned as a current possibility for controllers?
(A) Only single-threaded applications are supported.
(B) Multi-threaded applications in controllers are also possible.
(C) Controllers are being replaced by switches for processing.
(D) Controllers are becoming entirely hardware-based.

Q186. Physically, how far can a controller and a switch be from each other in an SDN network?
(A) Always one hop away.
(B) Always directly connected.
(C) Multiple hops away.
(D) Only within the same rack.

Q187. In a hierarchical or tree architecture for SDN controllers, how are controllers typically connected to switches?
(A) In a flat, one-to-one manner.
(B) In a ring-like fashion.
(C) In a tree-like fashion.
(D) In a fully meshed topology.

Q188. A key characteristic of the ring architecture for controllers, as described, is that a switch sends its packet-in request to:
(A) Any controller in the ring.
(B) All controllers in the ring simultaneously.
(C) A single, specific controller in the ring.
(D) The nearest controller in the ring.

Q189. What defines a centralized control mechanism in SDN?
(A) Control decisions are distributed among multiple controllers.
(B) A single controller manages the entire network.
(C) Switches make independent control decisions.
(D) Applications directly control the data plane.

Q190. What does a backup controller typically maintain to ensure uninterrupted network management?
(A) A separate, independent configuration.
(B) A replica of the main controller's state.
(C) Only a list of active switches.
(D) A log of all data plane traffic.

Q191. The example of security policy enforcement in SDN involved a 'policy chain' consisting of which components?
(A) Router, Switch, Server.
(B) Firewall, IDS, Proxy.
(C) Load Balancer, VPN, DNS.
(D) NAT, DHCP, ARP.

Q192. For SDN experimentation, what is a mandatory requirement for infrastructure deployment?
(A) It must be entirely virtualized.
(B) It must support OpenFlow.
(C) It must use proprietary hardware.
(D) It must be located in a data center.

Q193. How does a remote controller communicate with switches in an SDN setup?
(A) Through direct physical cable connections only.
(B) Using IP address and port number.
(C) Via wireless signals exclusively.
(D) By sending physical control signals.

Q194. What is characteristic of a local controller in an SDN environment?
(A) It is always located in a different geographical region.
(B) It communicates via IP address and port number.
(C) The controller itself is local and takes care of things locally.
(D) It only manages a single switch.

Q195. Given the trade-off of increased control packets, what is a current area of significant research in SDN?
(A) Maximizing the number of control packets.
(B) Minimizing the control logic and associated delay.
(C) Eliminating the need for controllers entirely.
(D) Standardizing proprietary control protocols.

Q196. The performance of SDN is primarily dependent on which two particular issues?
(A) Network bandwidth and latency.
(B) Rule placement and controller placement.
(C) Application layer protocols and transport layer protocols.
(D) Physical cabling and power supply.

Q197. What is an example of a function typically handled by the data plane in SDN?
(A) Defining flow rules.
(B) Forwarding data.
(C) Managing security policies.
(D) Configuring controllers.

Q198. Which components are part of the OpenFlow protocol as mentioned in the recap?
(A) HTTP requests and TCP connections.
(B) Flow rule and match fields.
(C) IP addresses and MAC addresses.
(D) Routing tables and ARP caches.

Q199. The East-Westbound API is relevant when the control layer consists of:
(A) A single, centralized controller.
(B) Multiple distributed controllers.
(C) Only data plane switches.
(D) Application servers.

Q200. When a switch sends a packet-in message to the controller, what specific information does the controller send back to the switch?
(A) The destination IP address of the packet.
(B) The instruction on how the switch is going to treat that particular flow.
(C) The current network topology map.
(D) A request for more packet-in messages.

Q201. What is a primary benefit of integrating Software Defined Networking (SDN) into Internet of Things (IoT) networks?
(A) Eliminating the need for physical devices.
(B) Enabling intelligent routing decision making.
(C) Restricting network resource visibility.
(D) Increasing the complexity of information collection.

Q202. How does SDN integration simplify network management in IoT?
(A) By requiring manual configuration for each device.
(B) By centralizing control over vendor-specific architectures.
(C) By simplifying information collection, analysis, and decision-making.
(D) By increasing the number of replicated data packets.

Q203. According to the provided IoT architecture reference model, which of the following is a distinct layer?
(A) Quantum Computing
(B) Data Abstraction
(C) Satellite Communication
(D) Biological Interface

Q204. The tiered segmentation of IoT networks includes which of the following tiers?
(A) Physical Tier
(B) Context-Aware Tier
(C) Security Tier
(D) Storage Tier

Q205. In an SDN-integrated IoT network, what is a key function of the SDN controller?
(A) To directly power IoT devices.
(B) To control different aspects and improve orchestration.
(C) To replace the need for a transport network.
(D) To perform packet segregation at gateways.

Q206. What type of control is made possible for IoT end devices (sensors, actuators, RFID tags) with SDN implementation?
(A) Distributed control
(B) Decentralized control
(C) Centralized control
(D) Autonomous control

Q207. Where is rule placement, considering issues like mobility and heterogeneity of end devices, typically handled in an SDN-integrated IoT network?
(A) At the data aggregator.
(B) At the access devices.
(C) Within the transport network.
(D) At the data center networks.

Q208. In an SDN-integrated IoT network, where are traffic engineering and backbone networks typically taken care of?
(A) At the IoT devices.
(B) At the data aggregator.
(C) At the transport network.
(D) At the gateways.

Q209. Which part of an SDN-integrated IoT network is responsible for flow classification and enhanced security?
(A) Mobile access channels.
(B) Data center networks.
(C) The SDN controller.
(D) The data aggregator.

Q210. What is a significant challenge regarding the programming of individual sensor nodes in a sensor network?
(A) Real-time programming is easily feasible.
(B) Real-time programming is typically not feasible.
(C) Sensor nodes are designed for complex programming.
(D) Programming is only done once during manufacturing.

Q211. What characteristic describes the architecture of sensor nodes from different vendors?
(A) Standardized architecture.
(B) Open-source architecture.
(C) Vendor-specific architecture.
(D) Cloud-based architecture.

Q212. What is a common limitation of sensor nodes regarding their computational capabilities and memory?
(A) They have unlimited computational power.
(B) They are resource-constrained.
(C) They possess vast amounts of memory.
(D) They can run any heavy computation.

Q213. The challenges in sensor networks present an opportunity to achieve which of the following?
(A) Reduce the number of sensor nodes.
(B) Program sensor nodes in real-time.
(C) Eliminate the need for data forwarding.
(D) Increase vendor-specific architectures.

Q214. The Sensor OpenFlow Protocol proposes which two types of data forwarding?
(A) Packet-centric and flow-centric.
(B) Value-centric and ID-centric.
(C) Real-time and batch processing.
(D) Unicast and multicast.

Q215. In value-centric data forwarding under the Sensor OpenFlow Protocol, when is data forwarded?
(A) When the source node ID matches a specific value.
(B) When the sensor data exceeds a certain threshold value.
(C) When the network traffic is low.
(D) When the destination node requests it.

Q216. How does ID-centric data forwarding work in the Sensor OpenFlow Protocol?
(A) Data is forwarded based on the type of sensor.
(B) Data is forwarded based on the ID of the source node.
(C) Data is forwarded based on the data's content.
(D) Data is forwarded based on the time of day.

Q217. What is a stated limitation of the Sensor OpenFlow Protocol?
(A) It only supports value-centric forwarding.
(B) Real-life implementation is not yet available.
(C) It requires excessive memory on sensor nodes.
(D) It is incompatible with SDN controllers.

Q218. Which of the following is a major component of the Soft WSN protocol?
(A) Cloud-based data storage.
(B) Sensor device management.
(C) Application layer security.
(D) Quantum encryption.

Q219. What significant aspect distinguishes the Soft WSN protocol's development compared to earlier solutions like Sensor OpenFlow?
(A) It introduced a new type of sensor.
(B) It performed real-life experimentation.
(C) It eliminated the need for a controller.
(D) It focused solely on theoretical models.

Q220. How does the Soft WSN protocol affect the packet delivery ratio compared to traditional WSN?
(A) It significantly decreases it.
(B) It has no impact on it.
(C) It much improves it.
(D) It makes it unpredictable.

Q221. What effect does Soft WSN have on the number of replicated data packets forwarded in the network?
(A) It significantly increases them.
(B) It has no effect.
(C) It reduces them.
(D) It only affects control packets.

Q222. Why does the Soft WSN protocol result in a higher number of control packets in the network compared to traditional WSN?
(A) To increase data packet replication.
(B) Because each node asks the controller for forwarding logic.
(C) To reduce overall network traffic.
(D) To bypass the controller entirely.

Q223. SDN-WISE proposes a platform for what kind of network?
(A) Software Defined Wide Area Network.
(B) Software Defined Wireless Sensor Network.
(C) Software Defined Wired Network.
(D) Software Defined Cloud Network.

Q224. In SDN-WISE, where is the flow table for rule placement available?
(A) At the central cloud server.
(B) At the SDN controller.
(C) At the sensor nodes.
(D) At the data aggregator.

Q225. How can nodes be programmed in real-time in SDN-WISE?
(A) Only through a proprietary language.
(B) Using any programming language through API.
(C) By physically reprogramming each node.
(D) It does not support real-time programming.

Q226. The sink component in the SDN-WISE architecture includes which two types of sinks?
(A) Primary and secondary.
(B) Real and simulated.
(C) Wired and wireless.
(D) Local and remote.

Q227. What protocol do the sensor nodes in SDN-WISE typically run?
(A) 802.11 Wi-Fi.
(B) Bluetooth Low Energy.
(C) 802.15.4.
(D) Ethernet.

Q228. What does INPP stand for in the context of SDN-WISE?
(A) Internet Protocol Packet Processing.
(B) In-Network Packet Progressing.
(C) Integrated Network Performance Protocol.
(D) Intelligent Node Power Management.

Q229. What is the overarching goal of integrating SDN into IoT networks, as described in the lecture?
(A) To replace all existing IoT infrastructure.
(B) To introduce more intelligence and improve network efficiency.
(C) To eliminate the need for human intervention.
(D) To standardize all vendor-specific architectures.

Q230. How does SDN integration affect the visibility of network resources and network management?
(A) It complicates visibility and management.
(B) It simplifies visibility and management.
(C) It has no effect on visibility but complicates management.
(D) It makes them entirely autonomous.

Q231. What kind of traffic analysis and decision-making is enabled by SDN in IoT?
(A) Manual traffic analysis and isolated decisions.
(B) Basic traffic monitoring and reactive decisions.
(C) Intelligent traffic pattern analysis and coordinated decisions.
(D) Random traffic generation and uncoordinated decisions.

Q232. In the simplified view of an IoT network, what component receives data from IoT devices via mobile or fixed access channels?
(A) The transport network.
(B) The data aggregator.
(C) The gateways.
(D) The SDN controller.

Q233. After data aggregation, where does the data typically pass through before reaching gateways for packet segregation?
(A) Directly to the application layer.
(B) A transport network.
(C) An edge computing layer.
(D) A physical device layer.

Q234. How does SDN implementation impact the service logic behind an IoT network?
(A) It complicates the service logic.
(B) It improves the service logic.
(C) It replaces the service logic entirely.
(D) It has no impact on service logic.

Q235. Due to resource constraints, what kind of programs are difficult to insert into sensor networks?
(A) Simple data collection programs.
(B) Basic communication protocols.
(C) Too many control programs.
(D) Firmware updates.

Q236. Besides real-time programming, what other opportunity arises from the challenges in sensor networks?
(A) To standardize all vendor architectures.
(B) To change the forwarding path in real-time.
(C) To eliminate the need for sensor nodes.
(D) To increase the computational power of each node.

Q237. In which publication was the Sensor OpenFlow protocol proposed by Low et al.?
(A) IEEE Systems Journal in 2016.
(B) IEEE INFOCOM Conference in 2015.
(C) IEEE Communication Letters in 2012.
(D) IEEE Internet of Things Journal in 2018.

Q238. The Soft WSN protocol, proposed by Bera et al., was published in which journal?
(A) IEEE Communication Letters.
(B) IEEE Systems Journal.
(C) IEEE INFOCOM.
(D) ACM Transactions on Sensor Networks.

Q239. Which of the following is a component of sensor device management within the Soft WSN protocol?
(A) Network routing optimization.
(B) Active sleep management.
(C) Data encryption.
(D) Cloud integration.

Q240. What is the primary role of topology management in the Soft WSN protocol?
(A) To manage application-layer protocols.
(B) To take care of node-specific management for data forwarding.
(C) To handle power supply for sensor nodes.
(D) To provide a user interface for the network.

Q241. In which conference was SDN-WISE published?
(A) IEEE Communication Letters.
(B) IEEE Systems Journal.
(C) IEEE INFOCOM.
(D) ACM SIGCOMM.

Q242. Besides sensor nodes and a sink, what other type of node is part of the SDN-WISE schematic?
(A) Gateway nodes.
(B) Controller nodes.
(C) Emulated nodes.
(D) Actuator nodes.

Q243. What is the base stack upon which SDN-WISE functions, according to the lecture?
(A) TCP/IP stack.
(B) 802.11 stack.
(C) 802.15.4 stack.
(D) Bluetooth stack.

Q244. What is a general conclusion regarding the use of SDN-based approaches for wireless sensor networks (WSN)?
(A) They complicate network management.
(B) They significantly improve network performance.
(C) They are only theoretical and lack practical application.
(D) They increase the cost of sensor nodes.

Q245. Beyond controlling individual aspects, what else does the SDN controller improve in an IoT network?
(A) The physical manufacturing process of devices.
(B) The orchestration between different devices and protocols.
(C) The power consumption of the entire network.
(D) The raw data acquisition speed.

Q246. In the IoT architecture reference model, which layer is responsible for processing data closer to the source?
(A) Data Accumulation.
(B) Application Collaboration.
(C) Edge Computing.
(D) Physical Devices.

Q247. What is the purpose of the 'data accumulation' layer in the IoT reference model?
(A) To display data to end-users.
(B) To store and manage collected data.
(C) To control physical devices.
(D) To provide network connectivity.

Q248. The lecture states that SDN can address a 'number of problems' facing the Internet of Things. What is the general nature of these problems?
(A) Primarily hardware manufacturing defects.
(B) Issues related to network management, efficiency, and intelligence.
(C) Legal and regulatory compliance issues.
(D) User interface design challenges.

Q249. What topic is mentioned as the focus for the next part of the lecture series?
(A) Advanced SDN controller design.
(B) How IoT devices like mobile devices can exploit SDN.
(C) The history of sensor networks.
(D) New sensor hardware technologies.

Q250. Which statement best summarizes the role of SDN in managing and controlling Wireless Sensor Networks (WSN)?
(A) SDN is irrelevant for WSN due to resource constraints.
(B) SDN is useful for managing and controlling WSN through applications.
(C) SDN replaces WSN entirely with traditional wired networks.
(D) SDN only provides theoretical benefits for WSN without practical implementation.

Answer Key

Q1: B) To integrate a DHT sensor and include networking components for remote data logging.
The lecture focuses on integrating a DHT sensor with Raspberry Pi and adding networking components to implement remote data logging using a client-server model.

Q2: C) DHT sensor
The lecture explicitly states, 'we are going to discuss the same integration we discussed previously that is integration of a dht sensor with raspberry pi'.

Q3: C) UDP
The lecture states, 'we are going to implement a python scripting for client making the raspberry pi behave as a client whereas on the remote desktop which will act as a server will use another python script which will act as a udp server'.

Q4: C) Client
The lecture specifies, 'making the raspberry pi behave as a client'.

Q5: C) Server
The lecture specifies, 'whereas on the remote desktop which will act as a server'.

Q6: B) To be obtained between the VCC and the data pin.
The lecture states, 'this four point seven kilo ohm resistor must be obtain between the bcc and the data pin that is between pin one and two'.

Q7: C) 4
The lecture mentions, 'from left to right if you keep the grid side up you have pins numbered one to four'.

Q8: B) Pin 2 (Data)
The lecture states, 'the second pin is used for transmitting data'.

Q9: C) Pin 11
The lecture states, 'we are connecting the data pin or pin two of the dht sensor to pin eleven of raspberry pi this pin eleven is in bcm mode'.

Q10: C) adafruit_dht
The lecture mentions, 'adafruit which supplies this dht twenty two sensor they also provide a library to integrate sensor to raspberry pi via python frame work so this has already been installed during our previous lecture now we are going to call the adafruit under score dht dot read retry function'.

Q11: B) To create an endpoint for network communication.
The lecture describes sockets as creating 'a two way communication between two nodes in a network' and that `socket.socket` is used to 'create a socket'.

Q12: B) `AF_INET`
The lecture states, 'socket family can be either af under square unix or af under square inet... for most of the general purpose programming we are going with af under score inet or internet protocols'.

Q13: C) `SOCK_DGRAM`
The lecture states, 'socket type can be either socket stream or socket data gram' and later, 'a socket data gram is being used for this particular program'.

Q14: B) The server's IP address and port number.
The lecture emphasizes, 'it is very important that each and every client should be aware of the servers ip address as well as the port to which the client is going to connect'.

Q15: B) BCM mode
The lecture states, 'this pin eleven is in bcm mode'.

Q16: C) 0
The lecture states, 'socket dot socket function it has three arguments first is socket family second is socket type and third is protocol which is by default zero'.

Q17: C) `bind()`
The lecture states, 'we ah put the host as socket dot get host name we assign a particular port and we bind to the port by calling the host name is the first argument and the port number as the second argument'.

Q18: B) The server sends a 'connection successful' message.
The lecture states, 'once your connection has been accepted it sends the server sends connection successful'.

Q19: B) To continuously listen for and receive data from clients.
The lecture states, 'now while it is true it will keep on infinitely looping the data and address will be received from the socket'.

Q20: C) It is written to a text file in append mode.
The lecture states, 'you open a any text file called data log text... you open a text file and whatever data you are receiving over the network you start writing to the text file... i have opened in append mode so that your data keeps on appending on to the same file again and again without overwriting the previous data'.

Q21: B) To ensure the server can uniquely identify and log data from each client.
The lecture explains, 'your clients do not need to have fixed ip addresses may be they can have a variety of ip addresses about which the server has no prior knowledge but it is very important that each and every client should be aware of the servers ip address as well as the port to which the client is going to connect... you need to know the exact location of each of these clients so that later on you can segregate the data and connect to it'.

Q22: B) The server will automatically create the file.
In the demonstration, the speaker deletes `data_log.txt` and then states, 'Now you see it has automatically created another data log dot txt'.

Q23: C) Humidity and temperature
The lecture consistently refers to the DHT sensor as a 'temperature and humidity sensor' and the readings as 'humidity and temperature'.

Q24: C) It is formatted as a string.
The lecture states, 'we formatted in the terms of a string and we transmitted over the socket'.

Q25: B) By relying on the IP address and port number received with the data.
The lecture explains, 'it will ah generate one address and this address can be stored and later on ah used for ah later on used for time stamping your data to which ah on your server in various data logs so imagine your your server has been connected to twenty different ip addresses or clients... you need to know the exact location of each of these clients so that later on you can segregate the data'.

Q26: C) The `close()` function is called to terminate it.
The lecture states, 'when this transmission is complete from ah from ah input receipt from the user and the user terminates the code the sock the close function will execute'.

Q27: C) 4096
The lecture states, 'data and address will be received from the socket and chunks of four zero nine six bytes'.

Q28: C) To add new data to the end of the file without deleting previous entries.
The lecture states, 'i have opened in append mode so that your data keeps on appending on to the same file again and again without overwriting the previous data'.

Q29: C) Lighting a match in front of the sensor.
The lecture states, 'for a demonstration i light up a match in front of the dht sensor and you can check the corresponding readings change over at the server'.

Q30: B) It stops receiving new data.
The lecture states, 'once this socket is close once i terminate this will send a message closing socket and closes down the socket you see the server code has stopped coming in right'.

Q31: B) To prevent warning messages from being displayed.
The line `GPIO.setwarnings(False)` is used to prevent warning messages from being displayed, as shown in the client code snippet.

Q32: B) BOARD mode
The client code snippet shown in the presentation explicitly includes the line `GPIO.setmode(GPIO.BOARD)`.

Q33: C) UDP is connectionless and generally faster for simple data streams.
The lecture describes UDP as 'data gram based connection' and focuses on one-way communication from client to server, which aligns with UDP's connectionless nature.

Q34: B) To retrieve the local machine's hostname.
The lecture states, 'we ah put the host as socket dot get host name'. This function returns the standard host name for the local machine.

Q35: C) The client's IP address and port.
The lecture mentions, 'you need to know the exact location of each of these clients so that later on you can segregate the data and connect to it'. The 'address' received with the data contains the client's IP and port.

Q36: B) It waits for connections from clients.
The lecture states, 'i start my server now as you can see the server has started it will wait for connections from the client and until then it will do nothing'.

Q37: B) Two numerical values (humidity, temperature) separated by a comma.
The lecture shows an example output: 'this seventy four point three and twenty five point seven so first one is for the humidity value and the second one is from the temperature readings'. This implies a comma-separated format.

Q38: B) The `adafruit_dht.read_retry` function is called repeatedly within a loop.
The client code snippet implies a loop where `sensor_data()` (which calls `adafruit_dht.read_retry`) is invoked repeatedly to get fresh readings before sending, ensuring new values are generated.

Q39: B) To convert the string message into bytes.
The transcript shows `sock.sendto(message.encode(), (server_ip, server_port))`, indicating that the string message is converted into bytes for network transmission.

Q40: B) The server will automatically resume receiving data from the client.
The lecture states, 'so whenever i start this code again my server will again start receiving this code'.

Q41: B) To build an interactive environment of networked devices.
The lecture states, 'basic aim of this internet of things is to create an interactive environment of network devices which are all connected together'.

Q42: B) Maintaining system performance and interactivity.
The lecture states, 'more the number of devices the more intelligent and interactive your whole system should be it should not be that you are increasing your device number of devices whereas your system performance is going down so by maintaining your system performance the device integration has to be made'.

Q43: B) To read temperature and humidity data from the sensor.
The lecture states, 'now we are going to call the adafruit under score dht dot read retry function to read data from the sensor'.

Q44: B) A server performs tasks requested by a client, while a client requests services.
The lecture states, 'the server performs the task or the service as requested by the client'.

Q45: C) Text file
The lecture states, 'you open a any text file called data log text... The demonstration then uses `data_log.txt`.

Q46: C) To associate the socket with the server's IP address and a specific port.
The lecture states, 'the address will be the servers address and the port will be the port which will be use for connecting various clients it and sock dot bind it starts the server port'.

Q47: B) It closes the socket connection.
The lecture states, 'when this transmission is complete... the user terminates the code the sock the close function will execute'.

Q48: B) To identify and segregate data from different clients in logs.
The lecture explains, 'you need to know the exact location of each of these clients so that later on you can segregate the data and connect to it'.

Q49: B) Data analysis, plotting, and refinement.
The lecture concludes with, 'in the next lecture we will be focusing more on what to do with the data and what exactly ah can be integrated can be done you can may be plot the data you can store the data you can refine the data you are storing and other such tasks'.

Q50: B) To define the numbering scheme for GPIO pins.
The client code snippet shows `GPIO.setmode(GPIO.BOARD)`, which sets the numbering scheme for the GPIO pins.

Q51: Server-side data processing and visualization.
The lecture explicitly states, 'in this part we will be focusing besides those previous applications will be focusing more on the server side that is once the data has arrived what exactly can be done with the data.'

Q52: To connect to sensors, read data, and send it to a remote server.
The system overview states, 'we will first start off with a network of temperature and humidity sensors connected with raspberry pi board... We read data from that sensor we send it to a remote server over the network received the data at the server.'

Q53: Pin 2
The lecture states, 'pin one is connected to power pin four to ground and pin two is connected to the data pin the raspberry pi interface is the same as before only changes pin two of dht is connected to pin eleven of the raspberry pi'.

Q54: To ensure data is properly formatted and handle corruption or incompleteness.
The lecture explains, 'there may be chances of data corruption there may be chances of ah incomplete data being transmitted... data may not be properly formatted and so on so in those cases you need to filter the data to create a seamless and well structured data base'.

Q55: To allow individual access to specific sensor values.
The lecture states, 'if we need to access individual data suppose you are getting temperature readings humidity maybe presence of gases light intensity and so on... for this to happen you need to split your data at the data log'.

Q56: split()
The lecture explicitly mentions, 'we will be using this basic inbuilt function of python called split so it can be used to split a string into multiple strings depending on the separator or the delimiter we are specifying'.

Q57: ','
The example given in the lecture is `data.split(',')` where the comma is the delimiter for splitting 'sunday,Monday,tuesday'.

Q58: matplotlib
The lecture states, 'regarding data plotting in python it is quite similar to plotting the data in matlab so both of these support a very powerful library called matplotlib'.

Q59: plot(x, y)
The lecture lists 'basic commands for plotting in two dimension are you write plot x comma y where you put the x axis and the y axis values'.

Q60: plt.ylabel()
The lecture demonstrates `myplot.y label` and later `plt.y lable` (with a typo in the transcript, but the context clearly points to `ylabel()`).

Q61: plt.ion()
The lecture lists 'ion is the interactive plot mode this is generally used for animations'.

Q62: coverage_plot()
The lecture states, 'on the server side we have added a new function called coverage plot what it does is it animatedly plots whatever data is in coming into the server over the socket'.

Q63: Humidity
The lecture explains, 'the data is being split in terms of two parts humidity and temperature so after splitting since the data was in a csv format so after splitting the zeroth element becomes humidity and the first element is temperature'.

Q64: fig.canvas.draw()
The lecture states, 'figure dot canvas dot draw it redraws the image without overwriting the previous parts'.

Q65: It only sends data, it does not receive.
The lecture mentions, 'the client doesnt require a ip address because we are aiming for a one way communication from the client to the server'.

Q66: It allows for easier and more efficient updates across multiple clients.
The lecture concludes, 'if you need to make changes make design your architecture in such a way that only changing the server will help you only changing the scripts on the server will help you change your whole implementation and functionality so i hope this was of some values to you'.

Q67: CSV
The lecture states, 'now imagine the data from the client is saved into a text file and the values are separated by a comma as we have covered in previous lectures this is called a csv or comma separated value file'.

Q68: Semicolon, colon, or even a question mark
The lecture says, 'instead of a comma you can use any other delimiter it may be a semicolon it maybe colon it maybe even a question mark better avoided but still'.

Q69: A list of individual substrings.
The example shows `data.split(',')` resulting in 'individual substrings sunday monday tuesday'.

Q70: To give a title to the entire plot.
The lecture states, 'title gives title to the whole plot'.

Q71: It uses the index of the values in 'a'.
The lecture explains, 'the x axis is if not provided it is automatically taken'. In the demonstration, `range(1,20)` for `a` results in a linear plot, implying the index is used for the x-axis.

Q72: numpy
The lecture states, 'we call a random function from numpy library module lets call it import numpy as np'.

Q73: plt.grid(True)
The lecture demonstrates, 'if you add this command it will show a it willturn up a grid on your plot'. The command shown is `plt.grid(True)`.

Q74: The server's IP address and the particular port it is listening on.
The lecture states, 'over here your server address is the server address and the particular port to which the server is listening'.

Q75: 4096 bytes
The lecture states, 'it is receiving data in four zero nine six bytes junks'.

Q76: To ensure the plot is displayed in a specific, existing figure window.
The lecture explains, 'you set the figure two similar number as number one otherwise it will keep on creating new windows every time it plots something new'.

Q77: In the form of h (humidity) and t (temperature).
The lecture states, 'we are packaging the data in the form of h and t and transmitting it over the socket to the server'.

Q78: Remotely actuating sensors or IoT nodes.
The lecture states, 'you may be able to remotely actuator sensors or actuators or ah iot node from any part of the world'.

Q79: To develop an interactive environment using a network of connected devices.
The lecture states, 'we are mainly focusing on creating an interactive environment using a network of connected devices'.

Q80: On both the client side and the server side.
The lecture states, 'the data may be processed on the client side as well as it may be processed on the server side itself'.

Q81: Power
The lecture states, 'pin one is connected to power'.

Q82: Ground
The lecture states, 'pin four to ground'.

Q83: Adafruit Python library
The lecture states, 'we already have the adafruit python library for this dht sensor'.

Q84: On the Raspberry Pi (client device).
The lecture states, 'you must remember this iot sr dot py file is kept on the raspberry pi'.

Q85: Client-server architecture.
The lecture states, 'now when we attempt to connect this over the network we create a client server architecture'.

Q86: Client sending too much data.
*The lecture mentions 'chances of data corruption', 'incomplete data being transmitted', 'server not able to receive the whole data', and 'data may not be properly formatted'. It does not state 'client sending too much data' as a *reason* for corruption, but rather the *result* of rapid uploads.*

Q87: By using the split() function with a comma as the delimiter.
The lecture states, 'the values which were incoming it were in the form of string h commastring t so we need to split these values on the server in the form of individual h and t s so we will be using this basic inbuilt function of python called split'.

Q88: To create a new figure (window) for plotting.
The lecture lists 'figure it creates a new figure'.

Q89: It adds a subplot within the current figure.
The lecture lists 'subplot it adds a subplot in the figure'.

Q90: scatter()
The lecture lists 'scatter makes a scatter point plot of the given points'.

Q91: The server is receiving data, storing it, and calling the plotting function.
The lecture describes, 'while it is true it is receiving data in four zero nine six bytes junks you open a data log file store the data to it and additionally call this plotting function'.

Q92: When the server is closed by the user or the client is closed.
The lecture states, 'once the server is closed by the user or the client is closed by the uses this plotting function will stop'.

Q93: Index 0: Humidity, Index 1: Temperature
The lecture states, 'the data is being split in terms of two parts humidity and temperature so after splitting since the data was in a csv format so after splitting the zeroth element becomes humidity and the first element is temperature'.

Q94: Temperature on Y-axis, Iteration on X-axis.
The lecture states, 'you plot the temperature versus the iteration that is ah normal temperature plot with respect to whatever data is incoming'.

Q95: Only the server script needs to be updated, saving time and effort.
The lecture states, 'suppose you have hundred clients youdo not need to update each and every client even if you need some minor adjustments make some minor adjustments so you can just update a singular source code or a singular script file on the server end and that will save you a lot of time as well as a lot of effort'.

Q96: A sensor device.
The lecture states, 'this will be on the client device itself sorry this will be on the sensor device itself which is the raspberry pi we cannot call it a client yet because it has not been connected to the network as of now'.

Q97: Ability to access sensor readings from any part of the world.
The lecture states, 'if this can be hosted on the internet you will be able to access your sensor readings sensor data... from any part of the world'.

Q98: In a log file.
The lecture states, 'successive data will be in coming into the server and get stored into a log file'.

Q99: To display the generated plot on the screen.
The lecture states, 'the show function shows the plot'.

Q100: The temperature points on the plot started rising.
The lecture states, 'will try increasing the temperature little bit and you can see the plot changing as you can see the temperature points now logged i raising see the temperature has risen'.

Q101: To transform or restructure existing network infrastructure for efficiency.
The lecture states, 'when we talk about sdn its about transforming or restructuring the existing network infrastructure and that can be done in an efficient manner.'

Q102: Switches only have a local view and forward traffic in a distributed manner.
The lecture explains, 'each of these switches they do not have the global view they route the traffic they forward the traffic in a distributed fashion they just know locally what you know what has to be done.'

Q103: Presence of a centralized controller for network management.
The lecture explicitly states, 'there is no centralized control that is feasible in these networks there is no centralize controller' as a limitation, making its presence a non-limitation.

Q104: To separate the application and operating system from the hardware in each switch.
The lecture states, 'the whole idea in sdn is to take care of the limitations by separating the application and operating system from the hardware so earlier in the earlier view what we had seen is the app os and hardware were hold or were all put together in each of these switches and here what we are doing is we are separating the application and operating system from the hardware in each of these switches.'

Q105: SANE
The lecture mentions, 'in the year two thousand six a stanford university team proposed a clean slate security architecture which was termed as sane to control the security policies in a centralized manner'.

Q106: 2008
The lecture states, 'in two thousand eight the idea of software defined network was originated with the open flow project'.

Q107: Forwarding packets.
The lecture describes the data plane as performing 'forwarding individually' and later states 'this control plane has been separated in sdn from this data plane which takes care of things like forwarding'.

Q108: Network Function Virtualization (NFV).
The lecture states, 'this is made possible they the concept of sdn is made possible through the process of network function virtualization nfv'.

Q109: Northbound API
The lecture defines, 'the northbound api is between the control plane and the application plane so this particular interface is known as the northbound api in sdn'.

Q110: OpenFlow
The lecture states, 'this is known as the southbound api and the current protocol supporting this southbound api is the open flow protocol'.

Q111: Rule placement problem and controller placement problem.
The lecture identifies, 'it is come to two main challenges when we want to implement sdn in any network so these are the rule placement problem and the controller placement problem'.

Q112: Routing tables.
The lecture explains, 'here we are talking about flow tables instead of routing tables in the concept of sdn'.

Q113: Ternary Content Addressable Memory (TCAM).
The lecture states, 'the size of their you know ah the size of these flow tables are such that they have to be stored in a specialized memory which are known as a tcam memory ternary content addressable memory'.

Q114: It is very fast but limited in size and expensive.
The lecture notes, 'these memories are very limited in size and so that only a limited number of rules can be inserted in this flow tables and so basically the flow tables are residing in this tcam memories and this tcammemories are very fast that means fast processing can be done... they are very expensive as well'.

Q115: Packet-in message.
The lecture describes, 'on receiving a particular request for which no flow rule is present in the switch so what happens is the switch basically sends a packet which is called the packet in'.

Q116: 3 to 5 milliseconds.
The lecture states, 'typically there is a delay of thee to five milliseconds involved in the new rule placements'.

Q117: How to define rules to minimize packet-in messages to the controller.
The lecture lists, 'how to define the rules so that less number of packet in messages are sent to the controller' as one of the challenges.

Q118: They are fixed within a particular version of OpenFlow.
The lecture clarifies, 'in a particular version of open flow this fields are going to be fixed so there are different matching fields source ip destination ip source port priority etcetera in a particular version of open flow and that these fields are fixed as i already mentioned'.

Q119: Hard timeout deletes all rules (e.g., switch reset), soft timeout deletes rules based on inactivity.
The lecture explains, 'all the rules are deleted from the switch at a particular hard time in the hard time out methodology this can be done by resetting the switch for example and in the soft time out methodology if no flow is received associated with a particular rule for a timeout interval of time then this particular rule is deleted'.

Q120: SDN is a technology/concept, and OpenFlow is a protocol used for communication between data and control planes.
The lecture explicitly states, 'sdn and open flow are not one and the same sdn is a technology or a concept open flow is a protocol that is used to communicate between the data plane and control plane so they are distinct'.

Q121: Open vSwitch
The lecture lists 'indigo linc pantou open vswitch' as 'some of the popular open flow switch software that are available currently to support open flow'.

Q122: Because there is no centralized control over the network.
The lecture states, 'if one switch becomes down the other switches can will become affected and there is no centralized way that we can address this particular problem'.

Q123: They are cost expensive, run thousands of lines of code, and require significant RAM.
The lecture mentions, 'each of these switches running thousands of lines of code they have each switch has millions of gates logic gates that are implemented ah requiring more than ten gb of ram they are cost expensive'.

Q124: To define the control logic and manage flow rules.
The lecture states, 'the whole idea is to separate the control logic from the hardware switches to define the control logic in a centralized manner' and 'this controller basically takes care of this flow rules defining the flow rules'.

Q125: Proprietary operating systems for each switch.
The lecture lists 'hardware switches we have controller applications flow rules and different application programming interfaces or the api' as components. The goal of SDN is to separate OS from hardware, moving away from individual proprietary OS per switch.

Q126: By using a centralized control to quickly reconfigure routes.
The lecture states, 'operation cost and delay that are caused due to link failure can be significantly minimized using sdn'. This is enabled by the centralized control's ability to manage the network.

Q127: It determines the order in which rules are evaluated.
In the example flow rule, it says 'if a packet arrives then it will check the priority'. Higher priority rules are checked first.

Q128: The specific fields in a packet header that are compared against the rule.
The example flow rule shows 'protocol has to be tcp and if the ip destination is this and the destination port is this' as conditions for a match.

Q129: The command to be executed if a packet matches the rule.
The example flow rule shows 'then the instruction that means the action that has to be taken is to forward to the particular port'.

Q130: Because it is very expensive.
The lecture states, 'this tcam memory they are very expensive as well they are very expensive so you cannot you know very easily you cannot scale up we cannot you know get as much memory as we want because of the cost for ness'.

Q131: How to define or place rules at switches considering limited TCAM availability.
The lecture mentions, 'one of the challenges is how to define or place the rules at the switches while considering the availability cam i already mentioned to you that tcam is very small in size and it is very expensive'.

Q132: A Layer 3 switch, which is used interchangeably with 'router'.
The lecturer clarifies, 'i would like to mention that in this particular lecture and let that means lecture on sdn and sdn iot i am going to use the terms switch and router interchangeably so when i mention about switch and basically talking about the l three or the layer three switch and so l three switch and router these terms are going to be used interchangeably'.

Q133: OSPF
The lecture states, 'let us say for switching or routing the protocol that is used is the ospf protocol'.

Q134: They forward traffic in a distributed manner without a global view.
The lecture states, 'these switches they forward the traffic in a distributed manner and they do not have the global view of the network'.

Q135: To provide a logical view of the network for performing functions like security and routing.
The lecture explains, 'this network function virtualization basically gives a logical view of the network and based on that these functions are performed'.

Q136: Security.
The lecture lists 'security routing ah traffic engineering and different other applications' as functionalities performed by the application plane.

Q137: A completely new security architecture designed from scratch.
The lecture mentions 'a clean slate security architecture which was termed as sane', implying a fresh start or new design.

Q138: The Open Network Foundation.
The lecture states, 'in march two thousand eleven the open network foundation was formed'.

Q139: Because it requires architectural changes and hardware/software support.
The lecture explains, 'it is not like you know we can go and we can change the existing network we can transform the existing switches to support sdn it is not so simple you know i has to be architected the hardware have to support the software have to support and so on'.

Q140: Switch applications are tightly coupled with hardware/OS, while application plane applications are decoupled and centralized.
The lecture describes traditional switches having 'hardware then we have the operating system and we have the applications that are running on top' as a 'skeletal view of the stack'. In SDN, the 'application plane' is separated and performs 'functionalities like security routing ah traffic engineering and different other applications' in a centralized manner, decoupled from individual switch hardware.

Q141: It means the field does not require any specific match (wildcard).
The lecture states, 'if the priority matches maybe in a star means that you know it is it it doesnt require any match'.

Q142: OpenFlow has different versions, and each version has a fixed set of matching fields.
The lecture mentions, 'open flow has different versions and they have different number of matching field but in a particular version of open flow this fields are going to be fixed'.

Q143: It provides a global view of the network and centralized control logic.
The lecture emphasizes that traditional networks lack centralized control and a global view, and SDN's solution is to centralize the control logic, implying a global view.

Q144: To provide a platform for network-wide control and management.
The diagram shows the control plane containing the 'network operating system' which is responsible for the centralized control logic.

Q145: Because SDN can make IoT networks more efficient.
The lecture states, 'the software defined networking... has lot of potential for use ah in internet of things for making iot efficient'.

Q146: It is integrated, with app and OS tightly coupled to hardware.
The lecture describes, 'what we had seen is the app os and hardware were hold or were all put together in each of these switches'.

Q147: Juniper
The lecture states, 'many industries for example all these routing ah router companies like juniper cisco etcetera they basically announced that they want to incorporate sdn into their switches'.

Q148: To determine how streams of packets are handled by switches.
The lecture explains, 'these flow rules are going to determine what is going to be done with the streams of packets that are arriving at the switches'.

Q149: The challenge of deciding where to locate controllers for optimal performance and resilience.
The lecture lists 'controller placement problem' as one of the two main challenges, implying its importance in deployment strategy, which typically involves optimal location for performance and resilience.

Q150: It allows for dynamic and flexible network management.
The lecture highlights that traditional networks lack dynamic configuration and centralized control, implying that SDN's centralized control provides these benefits.

Q151: B) OpenFlow
The OpenFlow protocol is the protocol that is used for the southbound API communication between the data plane and the control plane.

Q152: C) Northbound API
The northbound API is used to communicate between the control layer and the application layer.

Q153: C) The use of multiple controllers in the control layer.
The East-Westbound API concept arises when talking about not a single controller but multiple controllers in the control layer.

Q154: B) Defining flow rules according to application-specific requirements.
Controllers basically define the flow rule according to the application specific requirements.

Q155: C) Approximately 200 requests per second.
Typically a controller can handle roughly about 200 requests in a second, applicable for single-threaded applications.

Q156: C) Network congestion with control packets.
If we have a small number of controllers for a large network, the network might be congested with control packets.

Q157: C) One hop distance.
The controllers are logically connected to the switches in a one hop distance.

Q158: C) Ring architecture
In the ring architecture, these controllers are placed in a ring-like fashion, but a particular switch is connected to only one controller in this version.

Q159: D) Mesh architecture
The mesh architecture basically increases improves the fault tolerance and improves the reliability of the network.

Q160: B) Centralized control
In the centralized mechanism, the control decisions are taken in a centralized manner, for example, a network is controlled by a single controller.

Q161: C) To take over control of the network if the primary controller fails.
If the primary controller is down, then the backup controller takes over. The backup controller has a replica of the main controller and controls the network to have uninterrupted network management.

Q162: B) HTTP -> Switch -> IDS -> Proxy -> Firewall -> Switch
The sequence of flow of instructions is like this: HTTP request comes, then it is sent from one switch to another switch, then it goes to the IDS, comes back, goes to the proxy and the firewall, and then finally to this particular switch and then out of the network.

Q163: B) Mininet
For switch deployment, the Mininet software is very useful; it is used to create a virtual network with OpenFlow enabled switches.

Q164: C) Both remote and local controllers.
Mininet runs on Python and supports remote and local controllers both.

Q165: C) Open Daylight and ONOS
Open Daylight and ONOS are the most popular ones that are used for controller configuration.

Q166: B) Emulators can be connected to real networks with real traffic, unlike simulators which simulate everything.
In emulators, the real traffic can flow through this emulator and the data can be analyzed, whereas simulators basically simulate the entire thing (packet flows, network, nodes).

Q167: B) To make the network more efficient.
The motivation behind SDN is to separate out the control logic from the data logic (data plane) to make the network efficient.

Q168: C) Increased control message overhead due to additional packets.
There is a trade-off as well, and we have seen that more number of control packets are going to flow through the network, leading to increased control message overhead.

Q169: B) Approximately 3 to 5 milliseconds.
Typically, in the rules, the placement of the rules if it is not already existing in the flow table of a switch typically takes about three to five milliseconds.

Q170: C) Minimize control-linked delay.
We do not want too much of a control delay control linked delay to happen there is a trade off definitely you know we cannot avoid this delay but we have to minimize this particular delay.

Q171: C) The control plane (controller).
In SDN, the control logic is basically taken care of by the control plane, meaning the controller.

Q172: B) Control layer and data layer.
The southbound API is between the control plane and the data plane.

Q173: C) All existing standard APIs.
For the northbound API, all these existing standard APIs that are used can still be used.

Q174: B) How the different flows and packets are going to be handled.
The controller knows how the different flows how the different packets are going to be handled.

Q175: B) Switch sends packet-in message to controller -> Controller sends flow rule back to switch.
The switch sends a packet in message to the controller if the switch already doesn't have this flow rule for the particular flow that it has received. It will send a packet in message to the controller and the controller is going to send back the flow rule corresponding to that.

Q176: B) Each sub-network is controlled by a different controller.
In the distributed mechanism, the control divisions can be taken in a distributed manner, for example, each sub network is controlled by a different controller.

Q177: B) Firewall, proxy, HTTP, and IDS.
One can have enhanced level of security in the network and in this particular case we will be taking help of the firewall proxy http etcetera etcetera and the ids.

Q178: C) To experiment and evaluate performance.
Experimentation can be done with the help of simulators or they can be done with the help of emulators to evaluate the performance.

Q179: C) Python
Mininet runs on Python.

Q180: C) Unified network management leveraging a global view.
What is required is to have an unified network management to be made ah should be made possible using sdn while leveraging the global view of the network.

Q181: C) Delay.
We have understood the rule placement problem and the tradeoff between tcam limitations limited memory tcam memory and the delay.

Q182: B) Optimizing the number and location of controllers in the network.
The other issue that is very important is the controller placement problem and this is what we are going to discuss next. This problem involves how and where we are going to place this controller.

Q183: B) All incoming requests from the switches.
The controller must be able to handle all these incoming requests from the switches.

Q184: B) Control overhead increases significantly.
If rule placement is not minimized, otherwise what is going to happen is there are there will be a huge control overhead.

Q185: B) Multi-threaded applications in controllers are also possible.
Currently multi threaded applications in controllers are also possible.

Q186: C) Multiple hops away.
Actually, physically when we are talking about the controller and the switch can be multiple hops away and typically they are multiple hops away.

Q187: C) In a tree-like fashion.
In the hierarchical or the tree architecture, controllers are placed and connected to these different switches in a tree like fashion.

Q188: C) A single, specific controller in the ring.
When the packet in request has to be sent, this packaging request will be sent to a single controller only not that it can be sent to any of the other controllers in the ring.

Q189: B) A single controller manages the entire network.
In the centralized mechanism, the control decisions are taken in a centralized manner, for example, a network is controlled by a single controller.

Q190: B) A replica of the main controller's state.
The backup controller has a replica of the main controller.

Q191: B) Firewall, IDS, Proxy.
It is an example of a potential data plane ambiguity to implement the policy chain this chain firewall ids proxy.

Q192: B) It must support OpenFlow.
The infrastructure deployment must be supported with OpenFlow.

Q193: B) Using IP address and port number.
In the remote basically the controller can be situated in a remote place and communicated using IP address and port number.

Q194: C) The controller itself is local and takes care of things locally.
A local controller basically does you know everything is local the controller itself is local and you know it takes care of it in a ah takes care of it locally.

Q195: B) Minimizing the control logic and associated delay.
More number of control packets are going to flow through the network but we want to minimize that control logic and this is where lot of research is being undertaken at present.

Q196: B) Rule placement and controller placement.
We have also seen that the performance of sdn basically depends on two particular issues the rule placement and the controller placement in the network.

Q197: B) Forwarding data.
The data plane takes care of functions like forwarding etcetera.

Q198: B) Flow rule and match fields.
We have also understood the open flow protocol and flow rule and the math rule ah math fields in the open flow protocol.

Q199: B) Multiple distributed controllers.
This terminology comes into picture when we are talking about not a single controller but multiple controllers in the control layer.

Q200: B) The instruction on how the switch is going to treat that particular flow.
The controller is going to send back the flow rule corresponding to that particular flow, that means how it is at how the switch is going to treat it, that particular instruction is going to be sent by the controller.

Q201: B) Enabling intelligent routing decision making.
The integration of SDN into IoT allows for intelligent routing decision making, which is a fundamental problem that can be solved.

Q202: C) By simplifying information collection, analysis, and decision-making.
SDN integration simplifies information collection, analysis, and decision-making processes in IoT.

Q203: B) Data Abstraction
The IoT architecture reference model includes layers such as physical devices and controllers, connectivity, edge computing, data accumulation, data abstraction, application collaboration, and processes.

Q204: B) Context-Aware Tier
The tiered segmentation of IoT networks includes the context-aware tier, network tier, and application tier.

Q205: B) To control different aspects and improve orchestration.
The SDN controller controls each of the different aspects and improves the orchestration between different devices and protocols in the network.

Q206: C) Centralized control
With the implementation of SDN, centralized control of IoT end devices, including sensors, actuators, and RFID tags, is made possible.

Q207: B) At the access devices.
Rule placement, while considering issues like mobility and the heterogeneity of end devices, can be implemented at the access devices.

Q208: C) At the transport network.
Traffic engineering and backbone networks can be taken care of at the transport network.

Q209: B) Data center networks.
Flow classification and enhanced security are taken care of at the data center networks.

Q210: B) Real-time programming is typically not feasible.
One of the challenges in sensor networks is that real-time programming of each sensor node is typically not feasible.

Q211: C) Vendor-specific architecture.
Sensor nodes and their corresponding networks often follow vendor-specific architectures, meaning there is no one standard.

Q212: B) They are resource-constrained.
Sensor nodes are resource-constrained, meaning they have limited memory and cannot perform heavy computations often required.

Q213: B) Program sensor nodes in real-time.
The challenges in sensor networks create opportunities, such as the ability to program sensor nodes in real-time.

Q214: B) Value-centric and ID-centric.
The Sensor OpenFlow Protocol takes care of forwarding in two different ways: value-centric data forwarding and ID-centric data forwarding.

Q215: B) When the sensor data exceeds a certain threshold value.
Value-centric data forwarding means that if the sensor data exceeds a certain threshold value, then the data is going to be forwarded.

Q216: B) Data is forwarded based on the ID of the source node.
ID-centric data forwarding means the data will be forwarded based on the ID of the source node if it is coming from a particular source node with a certain ID.

Q217: B) Real-life implementation is not yet available.
A limitation of the Sensor OpenFlow Protocol is that real-life implementation of such a method is not yet available.

Q218: B) Sensor device management.
The Soft WSN protocol adopts different component solutions, including sensor device management and topology management.

Q219: B) It performed real-life experimentation.
The Soft WSN protocol performed real-life experimentation and showed experimental results.

Q220: C) It much improves it.
The Soft WSN protocol shows a much improved packet delivery ratio compared to traditional WSN.

Q221: C) It reduces them.
The Soft WSN protocol reduces the number of replications of data packets over the traditional WSN.

Q222: B) Because each node asks the controller for forwarding logic.
The number of control packets in the network is higher using Soft WSN because each time a node receives a new packet, it asks the controller for adequate forwarding logic.

Q223: B) Software Defined Wireless Sensor Network.
SDN-WISE basically proposes a software defined WSN sensor network platform.

Q224: C) At the sensor nodes.
In SDN-WISE, the flow table for rule placement is available at the sensor nodes.

Q225: B) Using any programming language through API.
In SDN-WISE, any programming language can be used through API to program the nodes in real time.

Q226: B) Real and simulated.
In the SDN-WISE architecture, the sink basically has both components: the real sink as well as the simulated sink.

Q227: C) 802.15.4.
These sensor nodes basically run the 802.15.4 protocol in their work.

Q228: B) In-Network Packet Progressing.
INPP stands for the in-network packet progressing processing layer.

Q229: B) To introduce more intelligence and improve network efficiency.
What is very important is intelligence, more intelligence is proved into the network and improving the efficiency of the network with the help of the SDN technology.

Q230: B) It simplifies visibility and management.
The visibility of network resources and network management is simplified based on user device and application-specific requirements.

Q231: C) Intelligent traffic pattern analysis and coordinated decisions.
SDN enables intelligent traffic pattern analysis and coordinated decisions through the controller.

Q232: B) The data aggregator.
Data from IoT devices through mobile or fixed access channels is acquired and transmitted to the data aggregator.

Q233: B) A transport network.
After data aggregation, it passes through a transport network and from the transport network it goes to the different gateways.

Q234: B) It improves the service logic.
Overall, SDN implementation is going to improve the service logic that is behind it.

Q235: C) Too many control programs.
Due to limited memory, too many control programs cannot be inserted into these networks.

Q236: B) To change the forwarding path in real-time.
Opportunities include being able to program sensor nodes in real time and changing the forwarding path in real time.

Q237: C) IEEE Communication Letters in 2012.
The Sensor OpenFlow protocol was published in the IEEE Communication Letters in 2012.

Q238: B) IEEE Systems Journal.
The Soft WSN protocol was published in the IEEE Systems Journal in 2016.

Q239: B) Active sleep management.
Sensor device management comprises sensor management, delay management, and active sleep management.

Q240: B) To take care of node-specific management for data forwarding.
Topology management basically takes care of node-specific management for forwarding of the data from their specific nodes.

Q241: C) IEEE INFOCOM.
SDN-WISE was published in the IEEE INFOCOM conference in 2015.

Q242: C) Emulated nodes.
The SDN-WISE schematic includes sensor nodes, a sink, and emulated nodes.

Q243: C) 802.15.4 stack.
There is a microcontroller unit and there is an 802.15.4 stack on top of which SDN-WISE basically functions.

Q244: B) They significantly improve network performance.
Network performance can be improved significantly using SDN-based approaches over traditional sensor network approaches.

Q245: B) The orchestration between different devices and protocols.
The SDN controller is going to improve the orchestration between the different devices and between the different protocols that are running.

Q246: C) Edge Computing.
The IoT architecture reference model includes an 'edge computing' layer.

Q247: B) To store and manage collected data.
The IoT architecture reference model includes a 'data accumulation' layer, implying storage and management of collected data.

Q248: B) Issues related to network management, efficiency, and intelligence.
The lecture states that 'lot of problems that are facing Internet of Things can be addressed with the help of SDN technology' and that 'more intelligence is proved into the network and improving the efficiency of the network'. This implies problems related to management, efficiency, and intelligence.

Q249: B) How IoT devices like mobile devices can exploit SDN.
In the next part of the lecture, we are going to talk about how IoT devices like mobile devices can exploit the use of SDN to make them efficient.

Q250: B) SDN is useful for managing and controlling WSN through applications.
The lecture summarizes that 'SDN is useful to manage and control the IoT network' and 'wireless sensor nodes and network can be controlled using SDN based applications'.